VivoQuest Tech Focus

We intend to hit on the exact **"Holy Grail" business model** for the future of value-based care. Shifting from a traditional employee model to an **AI-enabled, asset-light marketplace** for home health is a massive, venture-scale idea.
By building a specialized marketplace of non-nurse clinicians (phlebotomists, social workers, nutritionists) and physical products (DME, health devices), we act as the **"Uber Health" for high-risk, homebound preventive care.**

🗺️ The Architecture of The Marketplace Model

Instead of employing a massive workforce, our proprietary AI agent coordinates a tripartite ecosystem:

1. The Providers (Supply Side)

- **The Labor Pool:** Instead of fighting the critical nurse shortage, we curate a marketplace of independent or contract-based specialists:
  - **Phlebotomists:** To close care gaps by drawing blood for A1C or lipid panels right at the kitchen table.
  - **Social Workers / CHWs:** To evaluate Social Determinants of Health (SDoH), such as food insecurity or fall hazards in the home.
  - **Physical Therapists & Dietitians:** To optimize post-discharge mobility and nutrition, preventing the physical decline that leads back to the ER.
- **The Supply Chain (Products/DME):** Seamless integration with local Durable Medical Equipment (DME) providers and health companies. If a worker spots a fall hazard, they click a button, and a shower bar or walker is dispatched via our platform.

2. The Payers (Demand Side)

- **Medicare Advantage & ACOs:** They seed the platform with their highest-risk, unmanaged members. They pay your marketplace because buying an in-home nutritionist consult or a $50 shower bar is vastly cheaper than absorbing a $16,300 hospital readmission.

3. The Matchmaker (Your AI Logistics Engine)

- Your AI doesn't just match by proximity. It optimizes for **clinical context, credentialing, and multi-stop bundling**. For example, if a patient needs an A1C blood draw *and* a home safety check, the AI automatically dispatches a dual-certified worker or bundles the visits to minimize travel overhead.

💎 Why This Model Wins

The core economic pillars that make it an incredibly attractive bet:

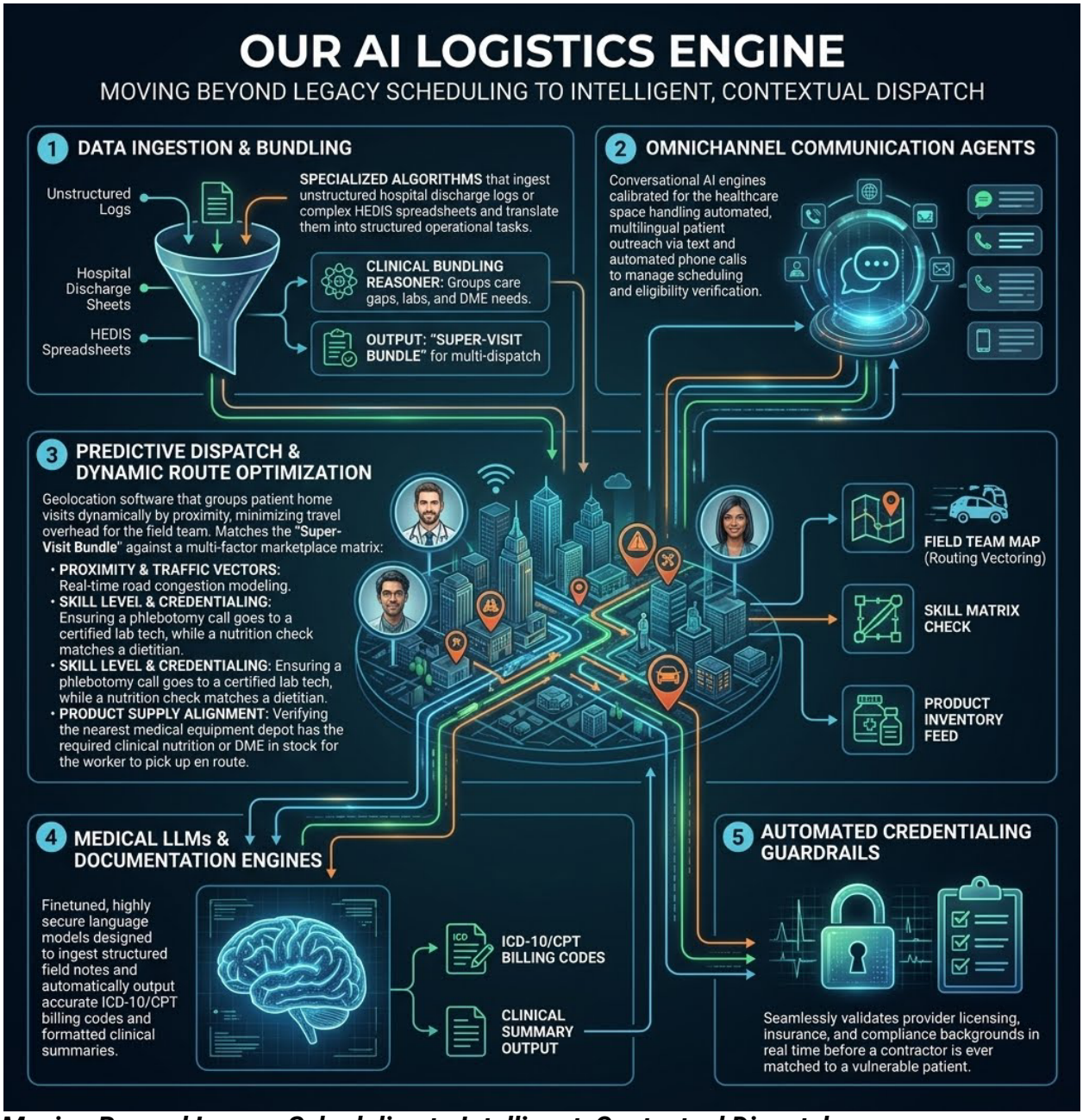
- **Negative Working Capital & Hyper-Scalability:** Traditional home health agencies grow by adding physical office footprints and salary overhead. We scale like Airbnb—matching underutilized local clinical talent with immediate payer demands, achieving a **fractional cost-per-visit**.
- **The "Closed-Loop" Advantage:** Existing tech competitors either send a chatbot *or* ship a box of medical equipment to the patient's house, hoping they figure it out. Our model delivers the **product AND the human hand** to set it up, ensuring 100% compliance for the health plan.
- **Unlocking New Payer Revenue Streams:** By utilizing social workers and nutritionists, we perfectly target CMS's **Supplemental Benefits** and **SDoH reimbursement codes**. Health plans get extra funding to pay for these exact services, meaning we are tapping into pre-allocated healthcare dollars.

The Solution

The "Uber Health" Marketplace for Preventative Care & Medical Goods

- **An Ecosystem Approach:** A two-sided marketplace that matches risk-bearing payers with a flexible, on-demand network of ancillary healthcare professionals (Phlebotomists, Social Workers, Nutritionists, and Physical Therapists) and Durable Medical Equipment (DME) suppliers.
- **Closed-Loop Delivery:** We don't just ship products or send digital alerts. We dispatch the right localized provider alongside the necessary equipment to ensure 100% compliance at the patient's kitchen table.
- **The Value Metric:** Shifting care from high-cost, scarce Registered Nurses to optimized, multi-disciplinary gig-economy providers cuts the total cost of preventative care delivery by over **50%**.

Our AI Logistics Engine:



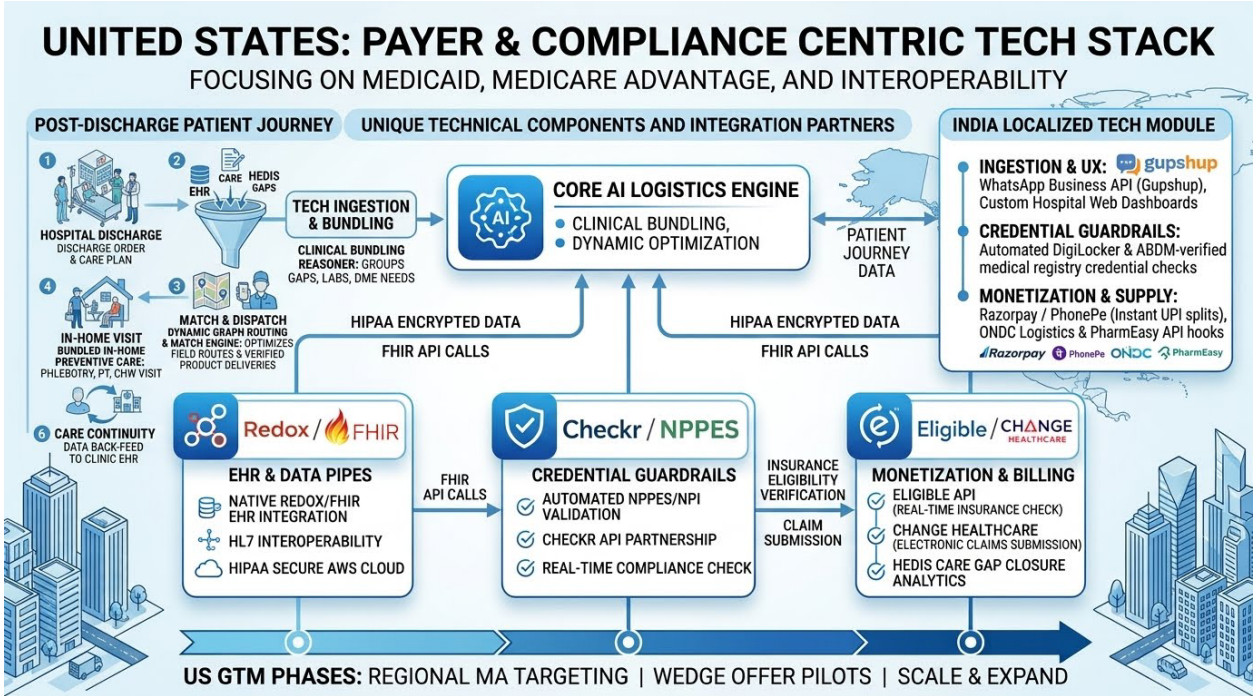
Moving Beyond Legacy Scheduling to Intelligent, Contextual Dispatch

The core software assets act as the central nervous system, automating administrative tasks so field workers can remain highly productive.

- **The AI Ingestion, Clinical Bundling, Orchestration & Logic Engine:** Specialized algorithms that ingest unstructured hospital discharge logs or complex HEDIS spreadsheets and translate them into structured operational tasks.
- **Omnichannel Communication Agents:** Conversational AI engines calibrated for the healthcare space that handle automated, multilingual patient outreach via text and automated phone calls to manage scheduling and eligibility verification.

- **Predictive Dispatch, Dynamic Route & Match Engine Optimization Software:** Geolocation software that groups patient home visits dynamically by proximity, minimizing travel overhead for the field team. Matches the "Super-Visit Bundle" against a multi-factor marketplace matrix:
    - *Proximity & Traffic Vectors:* Real-time road congestion modeling.
    - *Skill Level & Credentialing:* Ensuring a phlebotomy call goes to a certified lab tech, while a nutrition check matches a dietitian.
    - *Product Supply Alignment:* Verifying the nearest medical equipment depot has the required clinical nutrition or DME in stock for the worker to pick up en route.
- **Clinical Bundling Analytics:** Our proprietary AI engine analyzes a patient's open care gaps. Instead of ordering independent visits, it bundles needs—dispatching a worker who can simultaneously perform an A1C lab draw, conduct an SDoH safety checklist, and deliver a physical walker. **Output:** It collapses multiple potential appointments into a single **"Super-Visit Bundle,"** minimizing the cost of field deployment.
- **Medical LLMs / Automated Documentation Engines:** Finetuned, highly secure language models designed to ingest structured field notes and automatically output accurate ICD-10/CPT billing codes and formatted clinical summaries.
- **Automated Credentialing Guardrails:** Seamlessly validates provider licensing, insurance, and compliance backgrounds in real time before a contractor is ever matched to a vulnerable patient.

🗺️ The Patient Flow Journey:



The architecture diagram highlights the continuous, **three-stage cycle** designed by our engineering team:

1. **Stage 1: Hospital/Clinic Discharge (The Trigger)**
    - The hospital or clinical network initiates a discharge plan.
    - Instead of losing track of the patient, an automated webhook triggers your **Ingestion API** (utilizing FHIR/Redox protocols in the US or custom web/CSV uploads in India), capturing data on open HEDIS care gaps or necessary post-chemo tracking metrics.
2. **Stage 2: Core AI Logistics Engine (The Processing Node)**
    - Your central engine executes the **Clinical Bundling Analytics** to stack routine diagnostics, vitals check-ins, and medical device delivery needs into a single optimization request.
    - The dynamic graph router matches this request based on proximity, credentials, and real-time medical goods inventory availability.
3. **Stage 3: Verified Home Care Delivery (The Closed Loop)**
    - The local marketplace professional (phlebotomist, clinical nutritionist, PT or pharmacy products/ medical equipment) is dispatched to the patient's residence.
    - The provider performs the required tests (such as a cancer test kit administration or an A1C draw) and completes the setup of medical food products or DME right at the patient's kitchen table.
    - Care notes and diagnostic markers are automatically piped back to the originating hospital's EMR/WhatsApp portal, creating an airtight feedback loop.